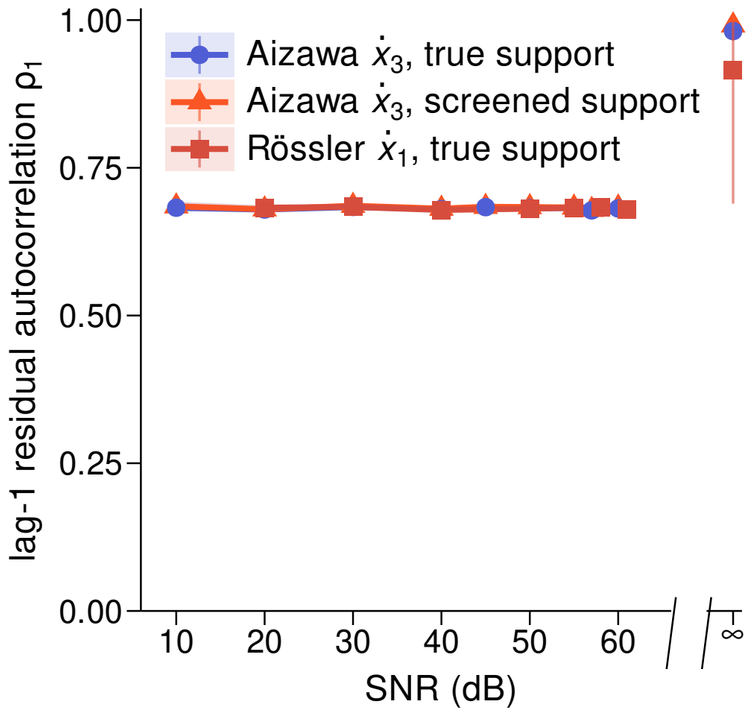
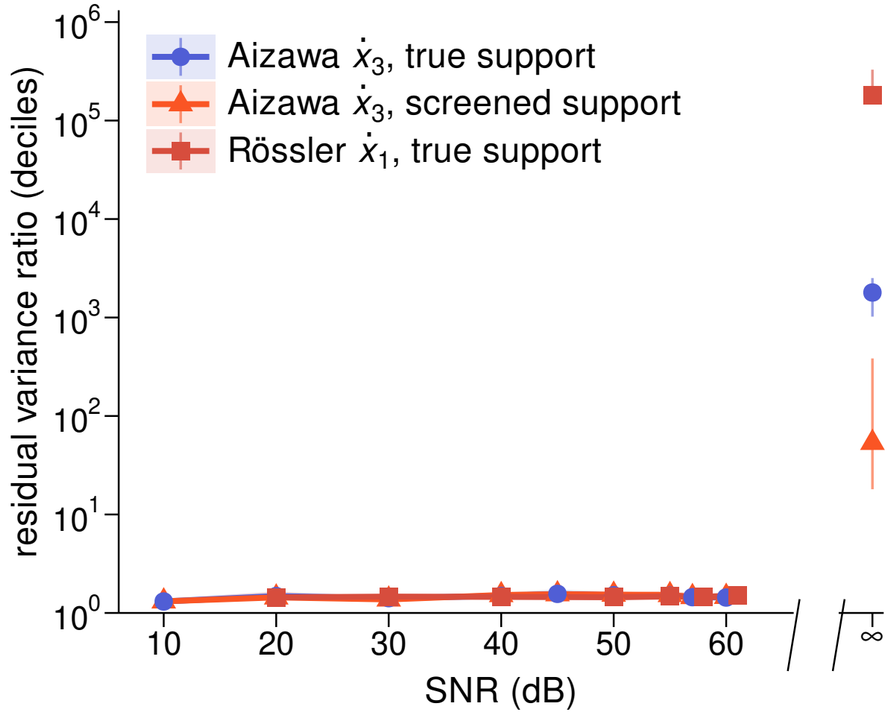
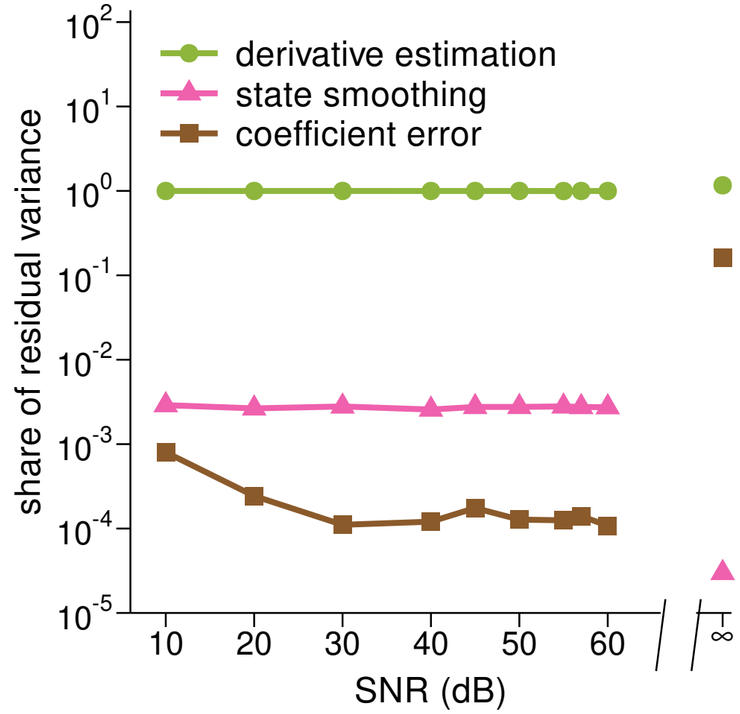
a


$n = 5000$
Aizawa $\dot{x}_3$, true support
Aizawa $\dot{x}_3$, screened support
Rössler $\dot{x}_1$, true support
lag-1 residual autocorrelation ρ_1
1.00
0.75
0.50
0.25
0.00
10
20
30
40
50
60
∞
SNR (dB)

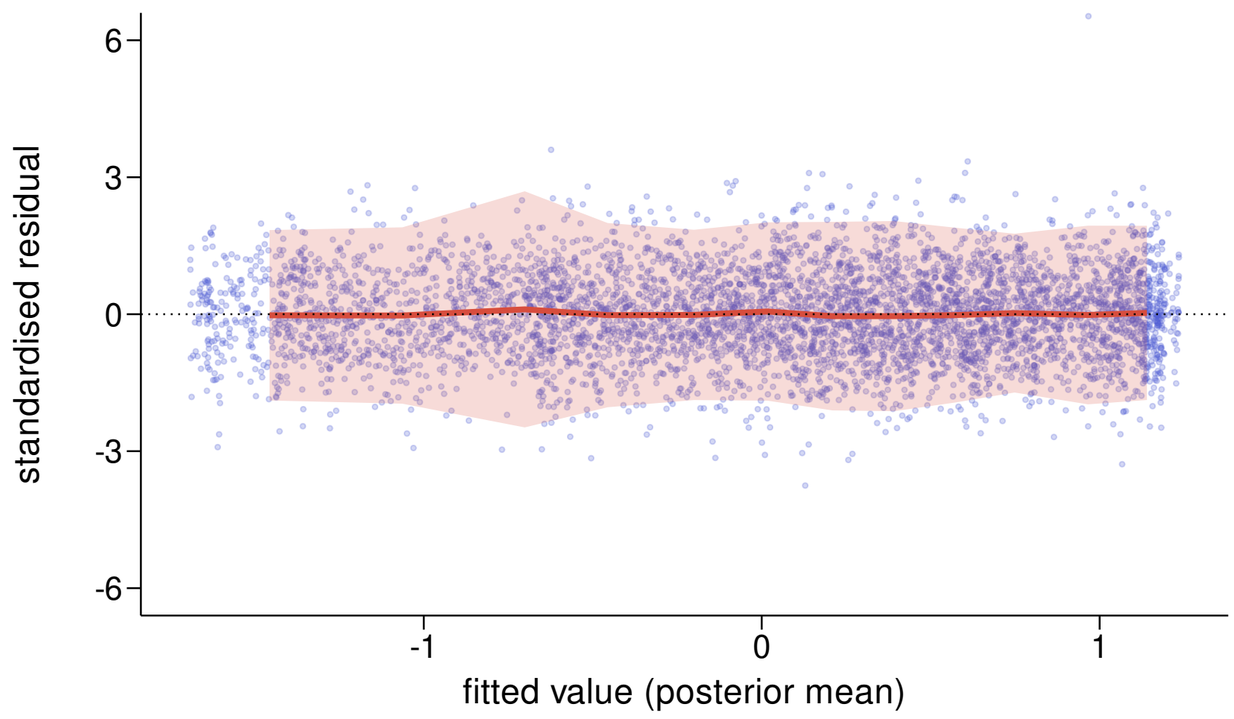
b


$n = 5000$
Aizawa $\dot{x}_3$, true support
Aizawa $\dot{x}_3$, screened support
Rössler $\dot{x}_1$, true support
residual variance ratio (deciles)
10^6
10^5
10^4
10^3
10^2
10^1
10^0
10
20
30
40
50
60
∞
SNR (dB)

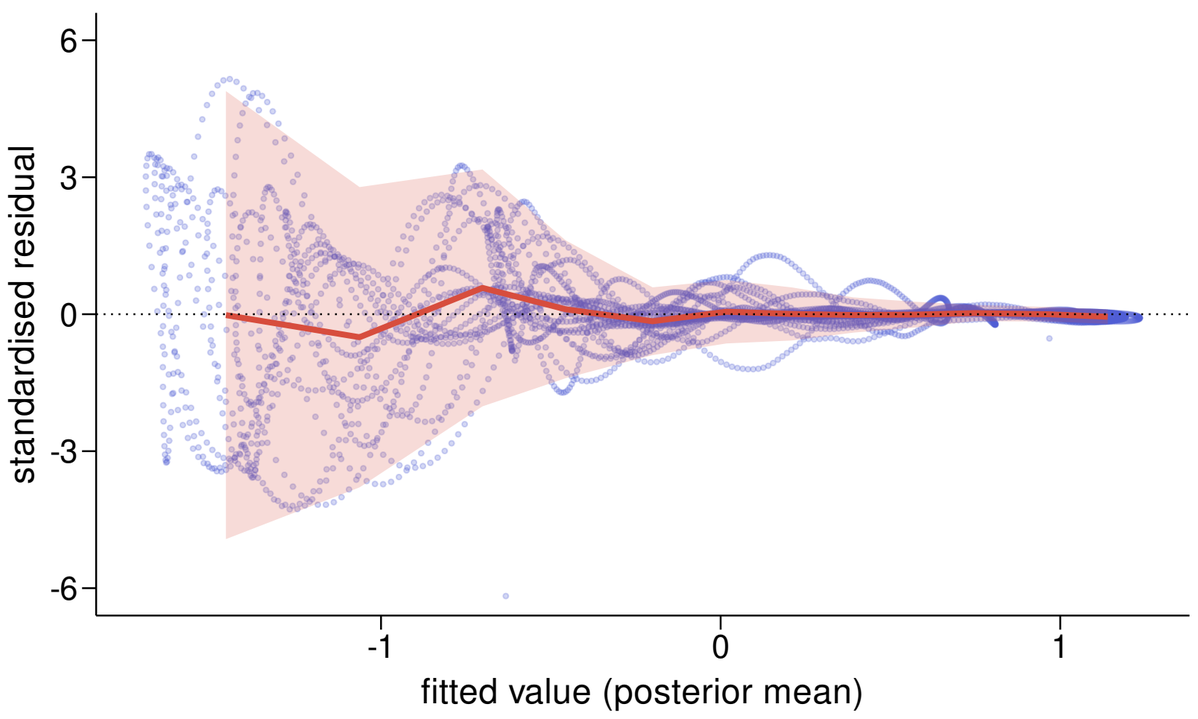
c


Aizawa $\dot{x}_3$, true support, $n = 5000$
derivative estimation
state smoothing
coefficient error
share of residual variance
10^2
10^1
10^0
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
10
20
30
40
50
60
∞
SNR (dB)

d


Aizawa $\dot{x}_3$, SNR = 60 dB
standardised residual
6
3
0
-3
-6
-1
0
1
fitted value (posterior mean)

e


Aizawa $\dot{x}_3$, SNR = ∞
standardised residual
6
3
0
-3
-6
-1
0
1
fitted value (posterior mean)